

ASP.NET Core Identity Learning Plan & Roadmap

Learning Objectives

Primary Goals

1. Master ASP.NET Core Identity implementation and customization
2. Understand Blazor Server architecture and best practices
3. Implement enterprise-grade authentication and authorization
4. Build a production-ready application demonstrating security features
5. Create a showcase product for stakeholder presentation

Secondary Goals

- Learn MudBlazor component library effectively
 - Understand SOAP/REST service integration
 - Master Entity Framework Core with Identity
 - Implement proper logging and monitoring
 - Deploy and configure production environment
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Phase 1: Foundation & Current State Analysis (Week 1-2)

Learning Objectives

- Understand existing codebase structure
- Master basic Identity concepts
- Review Blazor Server fundamentals

Tasks

1. Code Review & Documentation

- ☐ Document current authentication flow
- ☐ Map all Identity-related components
- ☐ Create architecture diagram
- ☐ Identify improvement areas

2. Identity Fundamentals

- ☐ Study Identity database schema
- ☐ Understand claims-based authentication
- ☐ Review password hashing mechanisms
- ☐ Learn about security stamps and tokens

3. Testing Current Features

- ☐ Test registration flow end-to-end
- ☐ Verify email confirmation process
- ☐ Test 2FA implementation
- ☐ Document any bugs or issues

Deliverables

- Technical documentation of current system
 - Bug report and improvement list
 - Initial project presentation outline
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Phase 2: Enhancement & Customization (Week 3-4)

Learning Objectives

- Implement custom Identity features
- Add role-based authorization
- Enhance user management

Tasks

1. Role Management System

- ☐ Create role management UI
- ☐ Implement role-based authorization
- ☐ Add role claims
- ☐ Create admin dashboard

2. User Profile Enhancement

- ☐ Add profile picture upload
- ☐ Implement department management
- ☐ Create user activity tracking
- ☐ Add user preferences system

3. Security Enhancements

- ☐ Implement account lockout policies
- ☐ Add password complexity rules
- ☐ Create login history tracking
- ☐ Add suspicious activity detection

Deliverables

- Fully functional role management system
- Enhanced user profiles with additional fields

- Security audit report
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Phase 3: Advanced Features (Week 5-6)

Learning Objectives

- Implement advanced authentication patterns
- Add social login providers
- Create audit logging system

Tasks

1. External Authentication

- ☐ Add Microsoft/Azure AD login
- ☐ Implement Google authentication
- ☐ Add LinkedIn provider (optional)
- ☐ Handle account linking

2. Audit System

- ☐ Create audit log database tables
- ☐ Implement automatic activity logging
- ☐ Build audit trail viewer
- ☐ Add compliance reporting

3. Advanced Security

- ☐ Implement API token authentication
- ☐ Add refresh token mechanism
- ☐ Create device management
- ☐ Add trusted device features

Deliverables

- Multi-provider authentication system
 - Complete audit trail functionality
 - API documentation
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Phase 4: Business Features (Week 7-8)

Learning Objectives

- Build business-specific features
- Implement workflow management
- Create reporting dashboards

Tasks

1. Organization Management

- ☐ Multi-tenant support (if applicable)
- ☐ Department hierarchy
- ☐ Team management features
- ☐ Organizational reporting

2. Notification System

- ☐ Real-time notifications with SignalR
- ☐ Email notification preferences
- ☐ Push notification support
- ☐ Notification templates

3. Dashboard & Analytics

- ☐ User activity dashboard
- ☐ Login statistics
- ☐ Security metrics
- ☐ Performance monitoring

Deliverables

- Business feature showcase
 - Analytics dashboard
 - Performance metrics report
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Phase 5: Polish & Production (Week 9-10)

Learning Objectives

- Optimize for production deployment
- Implement monitoring and logging
- Create deployment documentation

Tasks

1. Performance Optimization

- ☐ Database query optimization
- ☐ Implement caching strategies
- ☐ Minimize bundle sizes
- ☐ Add lazy loading

2. Production Preparation

- ☐ Environment configuration

- ☐ Secrets management
- ☐ Backup strategies
- ☐ Disaster recovery plan

3. Documentation & Training

- ☐ User manual creation
- ☐ Admin guide
- ☐ API documentation
- ☐ Video demonstrations

Deliverables

- Production-ready application
 - Complete documentation package
 - Deployment guide
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Showcase Features for Management

Core Demonstrations

1. Security Dashboard

- Real-time user activity
- Failed login attempts
- Security threat indicators
- Compliance metrics

2. User Management Portal

- Self-service capabilities
- Role administration
- Bulk user operations
- Import/export functionality

3. Business Value Metrics

- Reduced password reset tickets
- Improved security posture
- User satisfaction scores
- System performance metrics

Presentation Materials

- Executive summary (2 pages)
- Technical architecture document

- Live demo script (15 minutes)
 - ROI calculations
 - Future roadmap
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Success Metrics

Technical Metrics

- 100% test coverage for authentication flows
- < 200ms average page load time
- Zero critical security vulnerabilities
- 99.9% uptime target

Business Metrics

- 50% reduction in password reset requests
- 90% user adoption rate for 2FA
- 100% compliance with security policies
- Positive stakeholder feedback

Learning Metrics

- Complete all Microsoft Learn paths for Identity
 - Pass authentication/authorization assessments
 - Create 3 technical blog posts
 - Mentor 1 team member on Identity
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Recommended Learning Resources

Microsoft Documentation

- [ASP.NET Core Identity Fundamentals](#)
- [Blazor Security & Authentication](#)
- [Entity Framework Core with Identity](#)

Video Courses

- Pluralsight: "ASP.NET Core Identity Deep Dive"
- YouTube: Nick Chapsas Identity Series
- Microsoft Learn: Security Learning Path

Books

- "Pro ASP.NET Core Identity" by Adam Freeman
- "Blazor in Action" by Chris Sainty

Community Resources

- Stack Overflow Identity Tag
 - r/dotnet Reddit community
 - ASP.NET Core Discord server
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Risk Mitigation

Technical Risks

- **Data Migration Issues:** Create rollback procedures
- **Performance Problems:** Implement monitoring early
- **Security Vulnerabilities:** Regular security audits
- **Integration Failures:** Comprehensive testing

Project Risks

- **Scope Creep:** Define clear boundaries
 - **Timeline Delays:** Build in buffer time
 - **Resource Constraints:** Prioritize MVP features
 - **Stakeholder Alignment:** Regular demo sessions
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Weekly Progress Tracking

Week 1-2 Checklist

- ☐ Environment setup complete
- ☐ Basic features tested
- ☐ Documentation started
- ☐ Learning resources gathered

Week 3-4 Checklist

- ☐ Role system implemented
- ☐ User profiles enhanced
- ☐ Security improvements deployed
- ☐ First demo prepared

Week 5-6 Checklist

- ☐ External auth working
- ☐ Audit system complete
- ☐ API tokens implemented
- ☐ Integration tests written

Week 7-8 Checklist

- ☐ Business features complete
- ☐ Dashboards functional
- ☐ Notifications working
- ☐ Performance optimized

Week 9-10 Checklist

- ☐ Production deployment ready
 - ☐ Documentation complete
 - ☐ Presentation prepared
 - ☐ Knowledge transfer done
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Final Deliverable Checklist

Application Features

- ☐ Complete authentication system
- ☐ Role-based authorization
- ☐ User management portal
- ☐ Audit logging
- ☐ Email notifications
- ☐ 2FA implementation
- ☐ External authentication
- ☐ Security dashboard
- ☐ Performance monitoring
- ☐ API documentation

Documentation

- ☐ User manual
- ☐ Administrator guide
- ☐ Developer documentation
- ☐ Deployment guide
- ☐ Security procedures
- ☐ Troubleshooting guide

Presentation Materials

- ☐ Executive summary
 - ☐ Technical presentation
 - ☐ Live demo environment
 - ☐ ROI analysis
 - ☐ Future roadmap
 - ☐ Risk assessment
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Tips for Success

1. **Start Small:** Focus on MVP features first
 2. **Document Everything:** Keep notes of decisions and learnings
 3. **Test Continuously:** Automate testing where possible
 4. **Seek Feedback:** Regular demos to stakeholders
 5. **Stay Current:** Follow ASP.NET Core updates
 6. **Network:** Join community discussions
 7. **Practice Security:** Always think "security first"
 8. **Measure Progress:** Track metrics weekly
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Certification Path (Optional)

Consider pursuing these certifications to validate your learning:

- Microsoft Certified: Azure Developer Associate
- Microsoft Certified: Security, Compliance, and Identity Fundamentals
- CompTIA Security+ (for security fundamentals)